B.Sc. Engg. Part I, Even Semester Examination 2022
Dept. of Computer Science And Engineering
Course Code: PHY 1211
Course Title: Basic Electricity and Electrical Circuits

Section B

1.

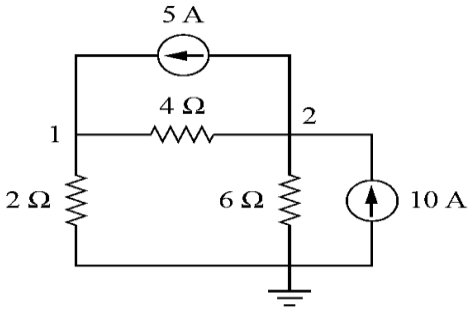


Figure A

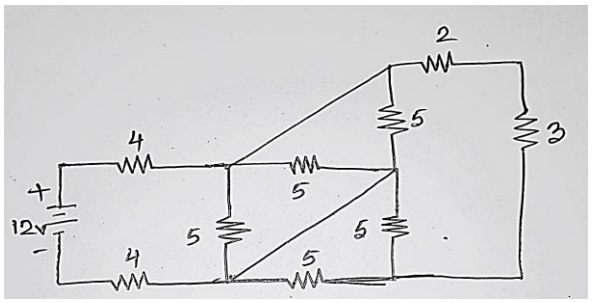
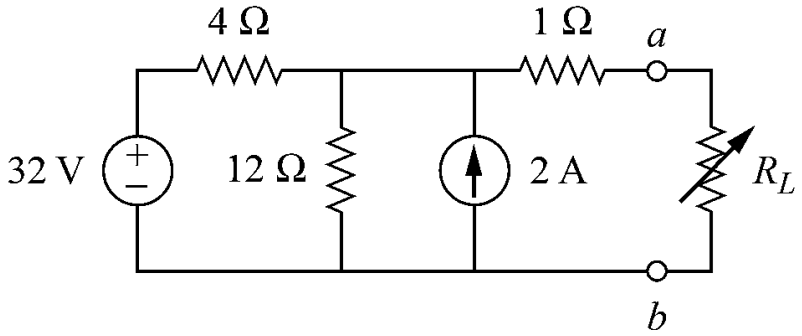


Figure |B

a) State and explain Norton's theorem. 3.00
b) Calculate the node voltages in the circuit shown in Figure A. 4.00
c) Find the equivalent resistance for circuit shown in figure B. Also calculate the return current of the independent source. All resistors are in ohm. 3.00

2.



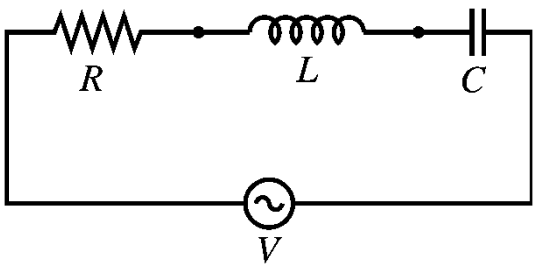
a) State and prove the maximum power transfer theorem for two terminal network. 3.00

b) State and explain the Kirchhoff''s current and voltage Laws. 3.00

c) Find the Thevenin equivalent circuit of the circuit shown in figure. Then find the current through R_L = 6 Ω 4.00

3.

When,
R = 25 Ω,
L = 10 mH
C = 250 μF
$V = 230\,V, 50\,Hz$

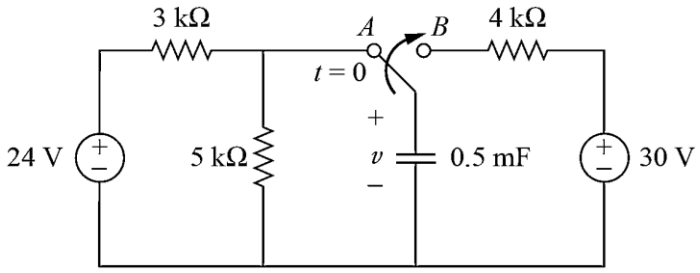


a) Define instantaneous voltage and RMS voltage. 2.00
b) Draw the timing and vector diagram for pure resistive, inductive and capacitive load. What is reactance and impedance of AC circuit? 3.00
c) Find Impedance, total current, voltage drop across each elements, draw voltage vector, power absorbed by resistance, power factor from 5.00

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4.



a) Write down the difference between steady and transient current 2.0

b) Define time constant for suitable example. 3.0

c) The switch in figure has been in position A for long time. At t = 0, the switch moves to B. Determine the time required for being v(t) = 25 V. 5.0